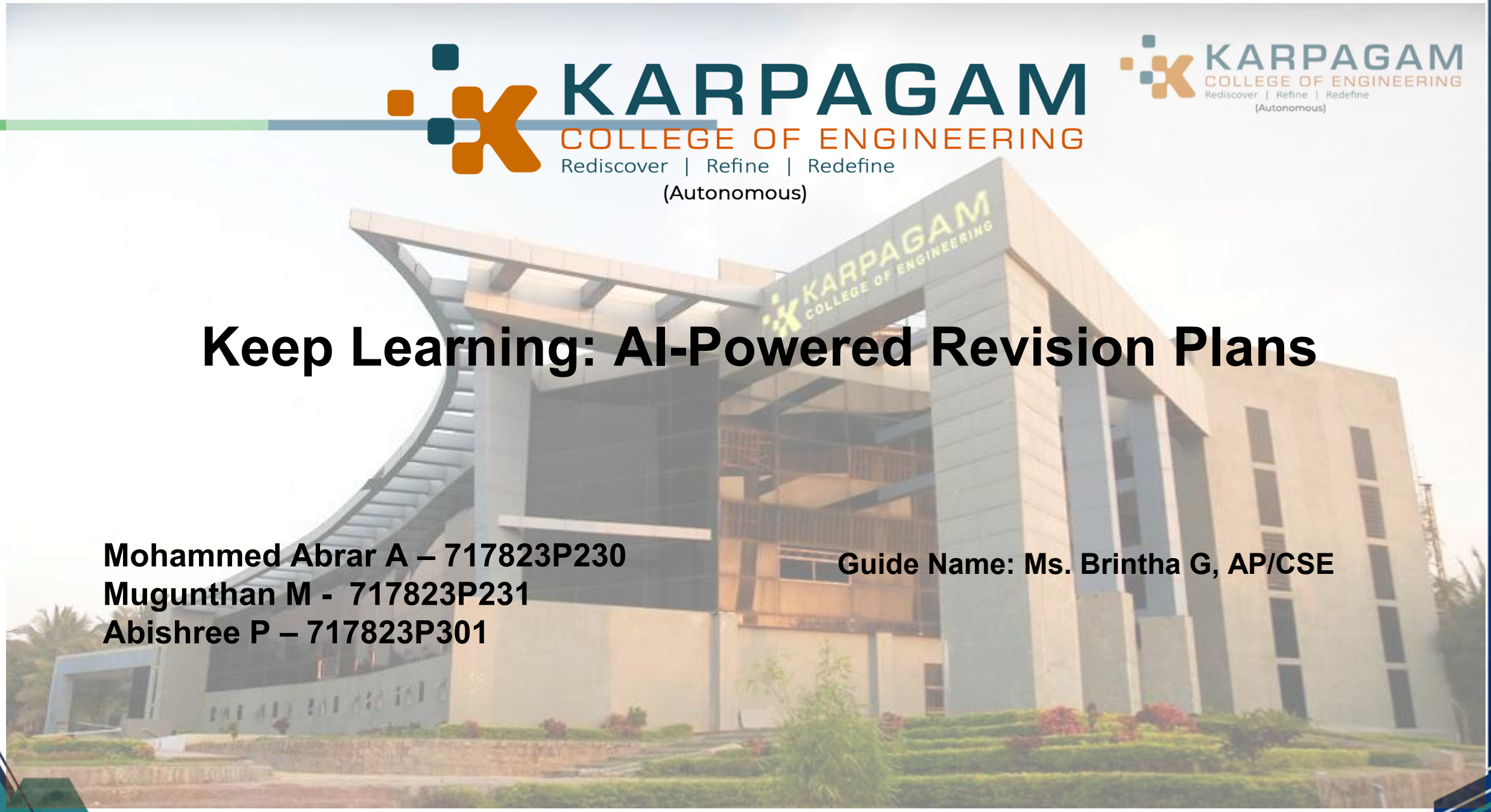
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Keep Learning: AI-Powered Revision Plans

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Introduction

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1. **Project Progress:** Review 2 converts the Knowledge Drift concept into a usable learning product through professional frontend design and structured application flow.

2. **Product Identity:** The system is presented as Keep Learning, an AI-powered platform that predicts knowledge loss and supports timely revision.

3. **Frontend Direction:** A responsive React-based interface is structured around dashboard analytics, topic-level knowledge tracking, AI insights, quizzes and revision planning.

4. **User Experience:** The complete learner journey runs from onboarding and learning to quiz results, prediction insights and personalized revision.

5. **Future Integration:** The frontend is built to connect later with the deep-learning prediction engine, backend APIs and database.

Problem Statement: Phase II

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1. **Implementation Gap:** The proposed prediction framework must become an understandable, usable learner-facing application.
2. **User Visibility:** Retention and knowledge-drift predictions must be shown clearly, not as raw model outputs.
3. **Personalization:** Topic-level risk must connect to a specific revision action, quiz and learning activity.
4. **Usability Challenge:** Learning, prediction, revision and analytics must live in one consistent, responsive application.
5. **Phase II Focus:** Build the complete frontend and clean interfaces ready for FastAPI, database and deep-learning integration.

Objective

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1 Professional Frontend

A responsive frontend for the Keep Learning platform.

2 Learner Dashboard

Knowledge score, retention, learning streak and topics at risk.

3 Knowledge Tracking

Subject-level and topic-level views with clear risk indicators.

4 Prediction Visualization

Current knowledge and predicted decline shown through interactive charts.

5 Quiz & Revision

Quiz, quiz-result and personalized revision-plan interfaces.

6 Analytics View

Knowledge growth, consistency, quiz performance and revision effectiveness.

7 Integration-Ready

Architecture prepared for backend and deep-learning integration.

Outcome of Review 2
A learner-facing product design, ready for real data and AI services.

Literature Review

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1. **Spaced Repetition Research:** Studies keep improving review scheduling using forgetting curves, half-life prediction and reinforcement learning.

2. **Knowledge Tracing:** LSTM-based and attention-based models represent learner knowledge and update it from sequential interactions.

3. **Forgetting-Aware Models:** Recent approaches incorporate time gaps and forgetting information to improve knowledge prediction.

4. **Project Relevance:** These studies support combining temporal learner history, retention prediction and personalized revision.

5. **Implementation Direction:** The literature guides the planned prediction logic and how risk and revision information is surfaced.

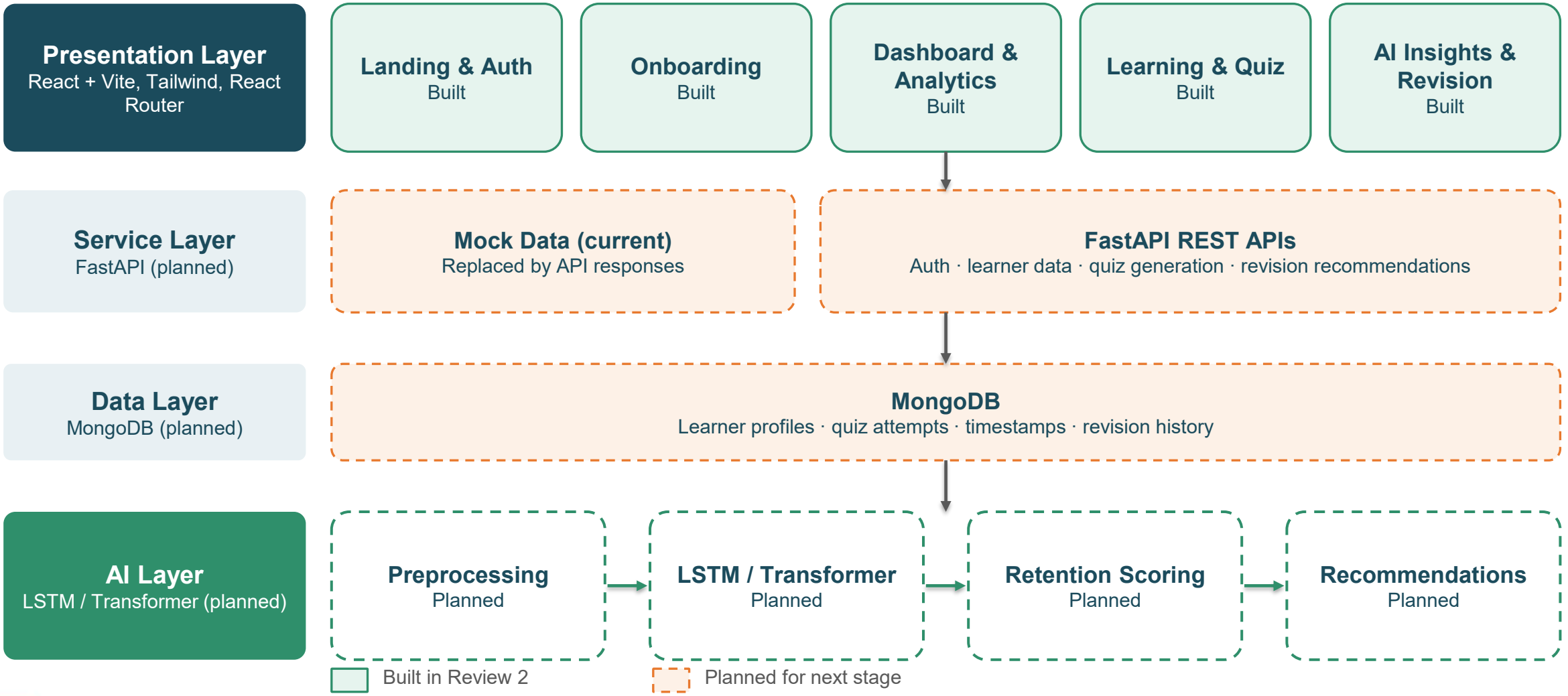
Methodology

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1. **Frontend Planning:** Define the learner journey, navigation structure, page hierarchy and reusable UI components.
2. **Mock Data Layer:** Create realistic subjects, topics, quiz results, retention values, predictions, revision tasks and history.
3. **Knowledge Visualization:** Represent knowledge score, retention, drift probability and predicted decline with cards, badges and charts.
4. **Learning & Quiz Flow:** Connect topic learning, practice questions, quiz submission and result analysis into one journey.
5. **Revision Recommendation UI:** Convert at-risk topics into prioritized revision tasks with time estimates and clear actions.
6. **Future Integration:** Keep data and UI boundaries ready for FastAPI APIs, MongoDB and LSTM/Transformer services.

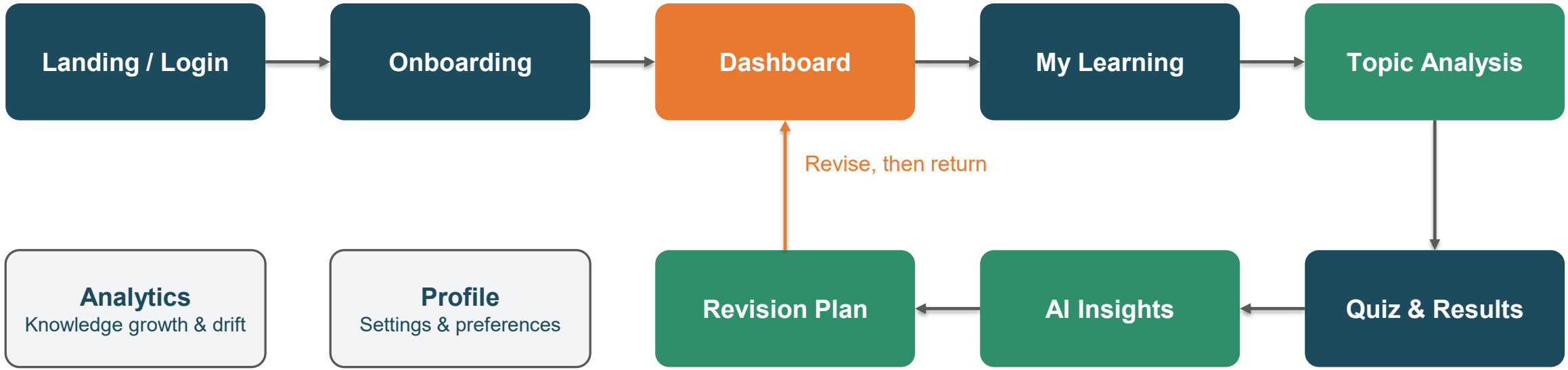
Proposed System Architecture

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Application Flow Architecture

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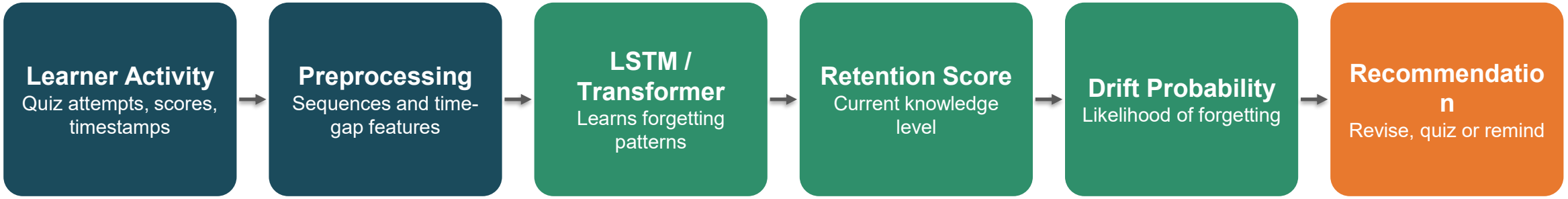
Entry: Learners sign in, complete onboarding and land on the dashboard.

Learning loop: From My Learning, a learner opens topic analysis, takes a quiz and reviews results.

Guidance: AI Insights explain predicted risk, and the Revision Plan schedules the topics to revisit.

AI Prediction Pipeline (Planned)

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Learner signals used

Quiz attempts and scores per topic

Timestamps and practice frequency

Revision history and topic-level performance

Why this matters

Predicting when a topic is likely to be forgotten lets the system recommend revision at the right time, not repeat a whole course.

The model output is translated into simple risk states and actions for the learner.

Learner Dashboard Design

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Knowledge Score
Overall mastery

Retention Level
How well knowledge holds

Learning Streak
Consistency over days

Topics at Risk
Items needing revision

Knowledge Trends

Score movement over time, so growth and decline are visible.

Subject Progress

Java, SQL, Data Structures and Spring Boot at a glance.

Recent Activity

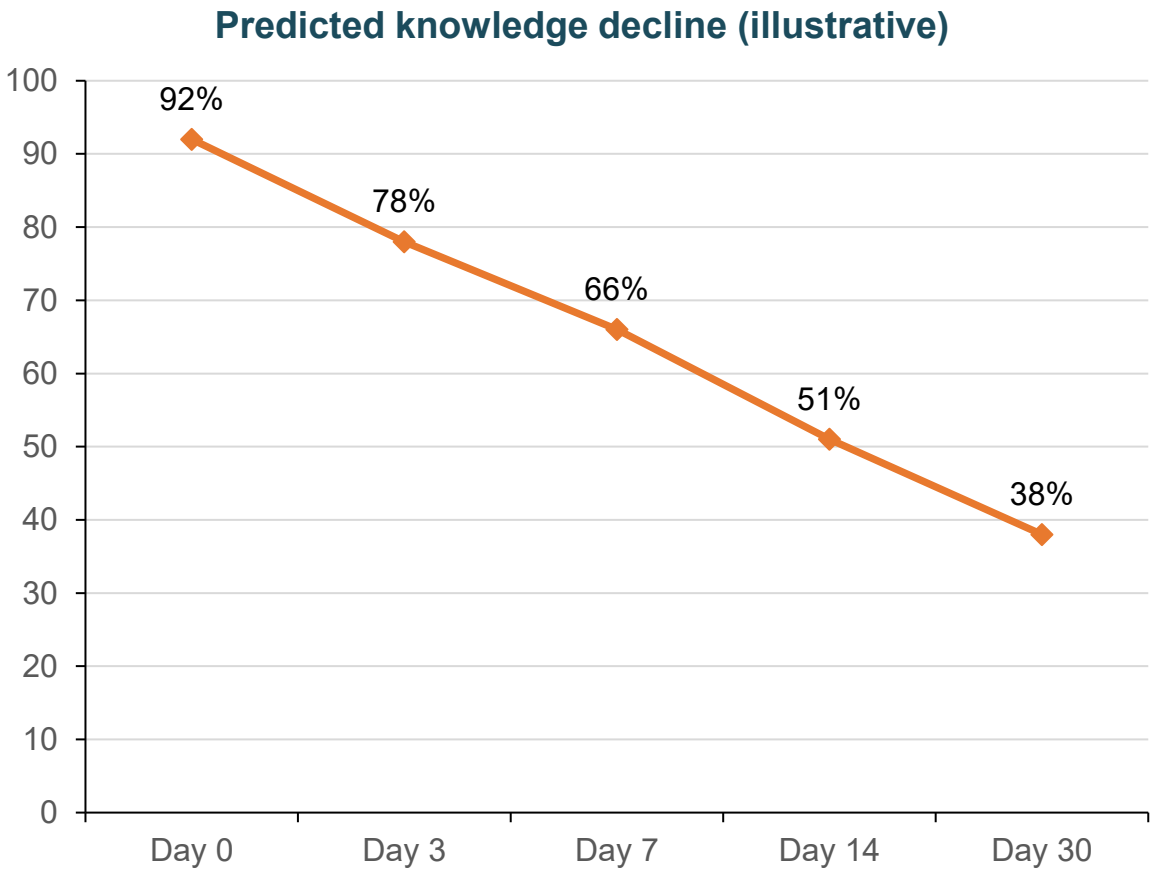
Latest quizzes, sessions and revision actions.

Today's Revision Priority

The topics to revise today, ranked by predicted risk.

Topic-Level Prediction View

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Current knowledge: estimated retention for the topic today

Drift probability: likelihood the learner forgets it soon

Stability: how firmly the topic is retained

Predicted decline: expected drop in knowledge over time

Risk factors: reasons behind the risk, such as long gaps

Recommended action: the revision, quiz or reminder to do next

Quiz, Revision & Analytics

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1 Quiz Module

- Topic-based assessments
- Concept-level performance analysis
- Result screen linking scores to weak concepts

2 Revision Plan

- At-risk topics grouped by priority
- Suggested revision time for each topic
- Clear actions for the learner

3 Analytics

- Knowledge growth over time
- Subject performance and quiz accuracy
- Learning consistency and drift trends
- Revision effectiveness

Technologies / Techniques Used

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1 React.js + Vite

Foundation for a modern single-page learning application.

2 Tailwind CSS

Consistent responsive design system and reusable styling.

3 React Router

Organizes the journey across landing, auth, learning and analytics.

4 Recharts

Knowledge trends, prediction forecasts and retention analytics.

5 Lucide Icons

Consistent icon system across navigation, cards and actions.

6 Mock State & Data

Realistic interactions before backend and AI services exist.

Implementation Results

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Completed in Review 2

Product identity: Keep Learning, with a clear knowledge-retention positioning.

Frontend architecture: Page structure, route hierarchy and responsive shell.

UI/UX design: Professional AI-SaaS / EdTech look with consistent cards and risk states.

Learner flow: Landing to onboarding, dashboard, learning, quiz, insights and revision.

Dashboard: KPI cards, knowledge trends, priorities and subject progress.

Prediction experience: Topic-level retention, drift probability, forecasts and actions.

Next Phase

Connect real data: Link the frontend with FastAPI and MongoDB.

Train the model: Build the LSTM/Transformer prediction engine.

Replace mock data: Swap mock predictions and learning data for real learner data.

Validate: Measure prediction performance on learner sequences.

End-to-end: Complete integration from learning activity to revision action.

Journal Base Paper Selection

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Base Paper
B. Settles and B. Meeder, "A Trainable Spaced Repetition Model for Language Learning," Proceedings of ACL.

1. **Core Contribution:** A trainable, data-driven approach to predict recall probability and memory half-life.
2. **Project Connection:** Predicting when a learner may forget an item supports the Knowledge Drift objective.
3. **Project Extension:** Keep Learning extends this from vocabulary items to academic and technical topics.
4. **Additional Layer:** Topic-level risk visualization, personalized revision planning, quizzes and analytics.

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[6] Liu Cheng Lee, Yutaka Arakawa, Tsunenori Mine, Emerging Trends in Knowledge Tracing Models: A Technical Survey from 2022 to 2025, Springer, 2025

[7] IEEE Xplore, Deep Learning Based Knowledge Tracing: A Review, a Tool and Empirical Studies, 2024

Conclusion

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1. **Progress:** Review 2 moved Keep Learning from a conceptual prediction framework to a complete learner-facing product design.
2. **Frontend Coverage:** Onboarding, dashboard, learning, topic analysis, quizzes, AI insights, revision planning and analytics.
3. **Core Idea:** Knowledge drift is made understandable through retention scores, risk indicators and predicted trends.
4. **Architecture:** A reusable, responsive frontend so backend and deep-learning services plug in without redesign.
5. **Next Stage:** Connect real learner data and the LSTM/Transformer engine, validate performance and complete integration.

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